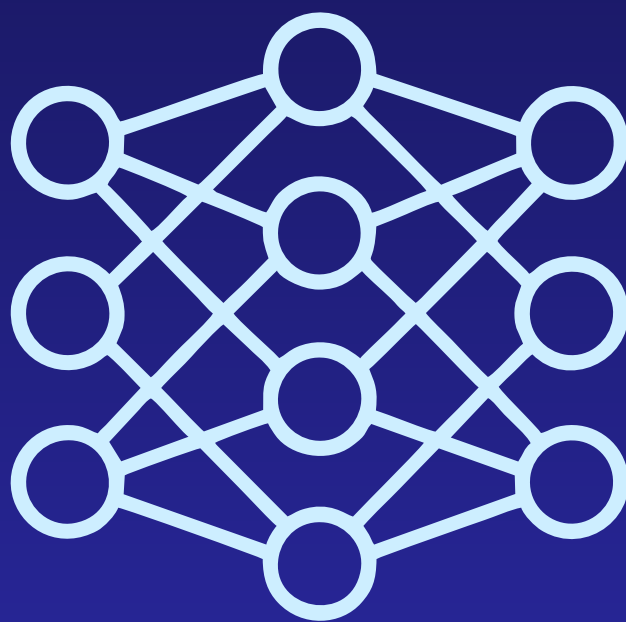




The 7 stages of the LLM Training Pipeline

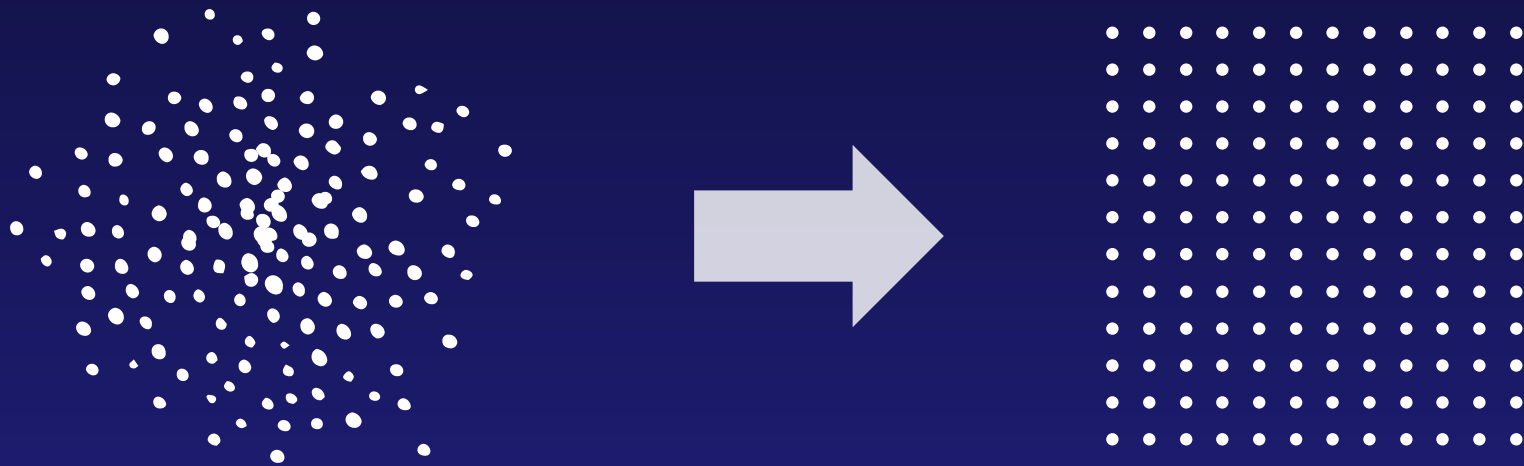


From raw data to
production-ready model



Pretraining

A model reads billions of tokens from the internet and learns the statistical patterns of language.



No instructions, no tasks, just raw next-token prediction.



This is where general knowledge is born.

1 / 7



Supervised Finetuning

The pretrained model knows language, but it doesn't know how to be useful.



SFT teaches it to follow instructions by training on curated input-output pairs.



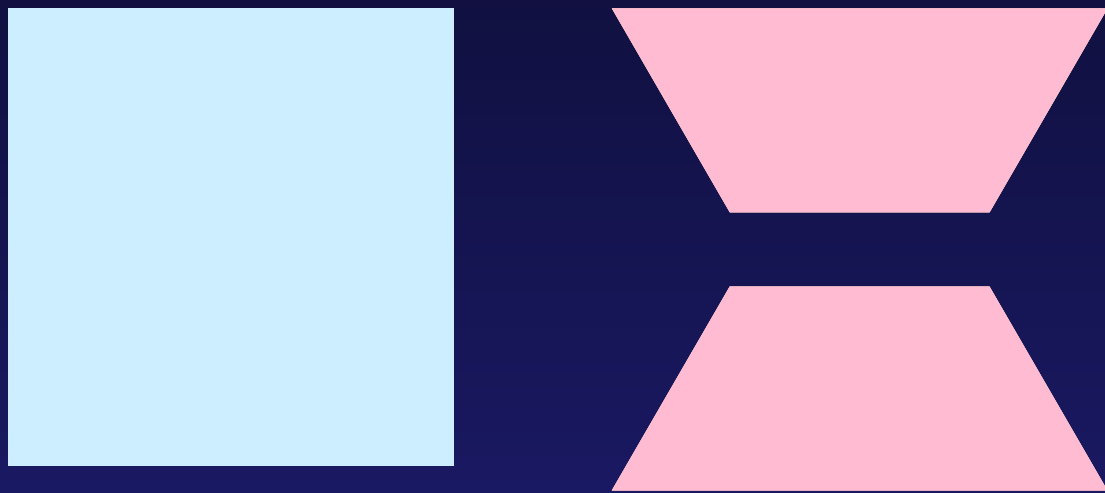
This is where a base model becomes an assistant.

2 / 7



LoRA

Finetuning all parameters of a large model is **expensive**.



LoRA freezes the original weights and injects small trainable matrices into key layers.



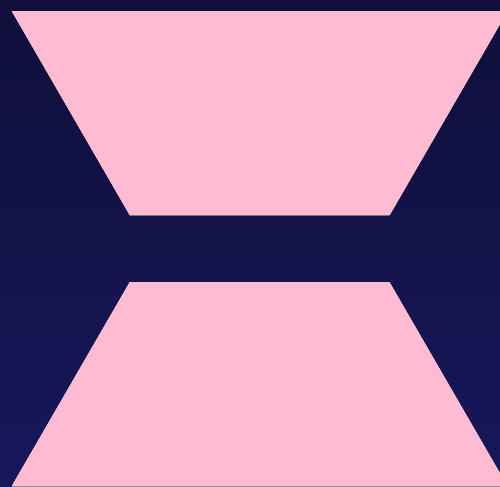
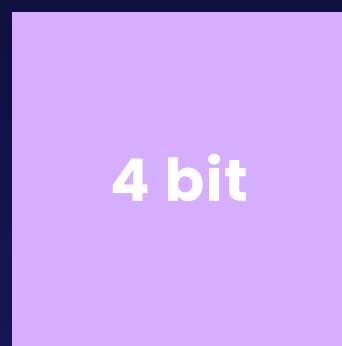
Same results, a fraction of the compute.

3 / 7



QLoRA

Take LoRA and push it further.



QLoRA loads the base model in 4-bit precision, slashing memory requirements.



Now you can finetune
a 7B model on a single
consumer GPU.

4 / 7



RLHF (PPO)

The model can follow instructions, but it might still be toxic, wrong or unhelpful.



RLHF trains a reward model on human preferences, then uses it to push the LLM toward responses humans actually prefer.



Now you have an aligned model.

5 / 7



Reasoning (GRPO)

GRPO skips the reward model entirely.



Instead, it samples multiple outputs, scores them with a verifiable rule, and reinforces the better ones.



This is how you teach a model to reason step by step.

6 / 7



Deployment

A model is only valuable when it's serving predictions.



Quantize it, wrap it in an API, optimize inference with vLLM, and ship it.



This is where training meets the real world.

7 / 7



Want to master the full pipeline?

Theory + Hands-on labs + Office
hours for **every stage**

Finetuning Sessions

theneuralmaze.substack.com

